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(1) Research Program/Project/Study Title: Ecological Profile of CapSU Pilar: Basis for Policy Crafting	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:		
Component Leader:		
Academic Rank:		

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	PILAR
Province:	CAPIZ
Region:	6

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☒ Basic	☐ Pilot Testing						
☐ Applied	☐ Tech. Promotion/Commercialization						

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(7) Sector/Commodity

(8) Discipline/Field of Specialization
Environmental Science

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Abstract

Everyone wants to go green – or, at least, they want their cities, municipalities, and public agencies to do so. Biodiversity drives various processes that make life possible by providing a number of ecosystem services. These includes regulations of biotic and abiotic factors and biological control. This study was conducted at Capiz State University, Pilar, Campus to assess its biodiversity as basis for policy crafting with the specific aim of 1. determining the surfaces, 2. assess the habitat quality, 3. determine the type of trees, 4. determine the understory and vegetation, 5. determine the environmental weeds, and 6. Identify the fauna present in the campus. The study employ the following methods: Four line transect using google earth photographs. Visual methods for identifying and enumerating the flora and fauna found in the area. Worksheet was prepared for gathering the data on the surfaces of the school, measuring trees and vegetation, enumerating the environmental weeds and monitoring of fauna. Journals and Plants portals were used in identifying the type of vegetation. Visual and monitoring were used to identify the habitat type. Results of the study revealed that much of the university ground were covered by soft surfaces and the quality of habitat assessment got a score of 67. There is a presence of native trees recorded and most of the vegetation cover are native. Most of the environmental weeds were indigenous and some actions were taken to remove it. There are also 10 species of fauna identified and recorded in the campus.

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Biodiversity drives various the processes that make life possible by providing a number of ecosystem services. These include the regulation of abiotic factors (climate, natural disturbance, gas, water, erosion control and sediment retention, soil formation, shelter, nutrient cycling, raw materials, waste treatment) and biotic factors (pollination, genetic resources, food production), and biological control. Biodiversity contributes significantly to the economy in a number of ways. It provides resources for research, tourism and the development of food and medicines. It can also increase agricultural production through the ecosystem services. It forms a fundamental part of values such as beauty and tranquillity. There are still Filipinos that place a high value on native plants and animals, which contribute to the sense of cultural identity, spiritual enrichment and recreation.

Understanding the quality and extent of biodiversity present in a given area is important for both land managers and the community. To effectively conserve and encourage biodiversity into areas you must first know the quality of what is there in the first place. It is therefore important that biodiversity is protected not only within nature conservation reserves but also on public and private lands.

Human activities are one of the most influential factors in bringing changes to the environment. These changes could be positive if actions are pro-environment such as reforestation and other cleaning and greening activities. On the other hand, effects could be abominable if the activities conducted can harm the environment. Each country should involve its people in protecting the environment and natural resources. Government and non government efforts are encouraged to sustain and save the environment, preventing drastic and worse resource deterioration.

Biodiversity conservation cannot be achieved with the efforts of one agency alone. This has to start from somewhere and as a government institution SUCs should be one the pioneer in biodiversity conservation of the country. Thus this study was conducted to assess the biodiversity of tCapSU Pilar Campus as basis for policy crafting towards conservation with the specific aim of 1. determining the surfaces within the campus, 2. assess the habitat quality, 3. determine the trees found in the campus, 4. determine the understory and vegetation within the campus, 5. determine the environmental weeds within the campus, 6. fauna present in the campus

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social

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Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Research Methodology

Study Area

This study was conducted in the campus of Capiz State University, Pilar. In identifying the study site, an aerial photograph using google earth was used. A grid was drawn in the picture covering it with smaller quadrat (m²) using the google tool measurement. Using the google earth the area of the ground is 3.536 hectares covering 48 boxes with a perimeter of 769.37 m.

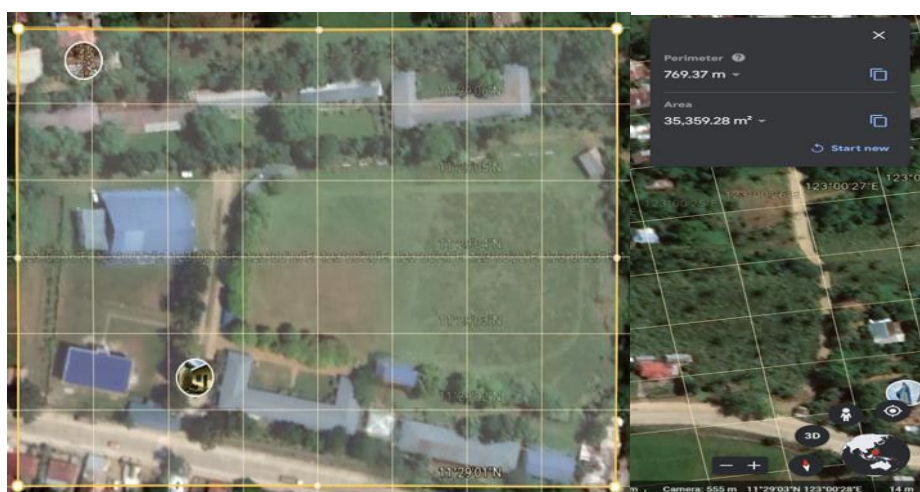


Figure 1. Aerial photograph of the study site

Gathering of Data

The university was surveyed using 4 line transect technique during the whole month of October 2022. Visual methods was used in identifying and enumerating the flora and fauna found within the campus. A worksheet was prepared for gathering data on the surfaces of the school, measuring trees and vegetation, enumerating the environmental weeds and monitoring the fauna. Journals and Plants portals were used in identifying the type of vegetation as to native (from the Philippines), indigenous (native and from the local) and exotic (introduce from other country). Visual and monitoring were use in the identification of habitat type weather it is habitable (with other organism) and not habitable (No organism is living).

Data analysis using

Biodiversity Audit Tool (Goyen, 2008) was used in analyzing the following:

1. In measuring the percentage cover of each surfaces the formula was used:

$$\text{Percent cover} = \frac{\text{Number of boxes covering each surfaces}}{\text{Number of hectares in the school gound}}$$

2. In measuring the number of trees/hectare

$$\text{No. of trees} = \frac{\text{Number of trees in the school ground}}{\text{No, of hectares in the school ground}}$$

3. In measuring the number of trees/hectare

$$\text{No. of habitat trees} = \frac{\text{Number of habitat trees in the school ground}}{\text{No. of hectares in the school ground}}$$

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No, of hectares in the school ground

4. In measuring percentage cover of understory

$$\text{Percent understory cover} = \frac{\text{Number of understory boxes shaded}}{\text{Number of boxes covering the school groundover of}} \times 100$$
5. In measuring cover of organic litter

$$\text{Cover of organic litter} = \frac{\text{Number of organic litter shaded}}{\text{Number of boxes covering soft surfaces}} \times 100$$
6. Inmmeasuring no. of rocks and logs/hectare

$$\text{No. of rocks and logs} = \frac{\text{No. of rocks and logs in the canmpus}}{\text{No. of hectares in the school campus}}$$

Scoring for Habitat Quality Assessment (Audit Tool Goyen, 2008)

The scoring and ranking for the habitat quality assessment were as follows:

No. of Trees

Variable	Score
<10 trees	0
10-20 trees	5
>20 trees	10

No. of Habitat Trees

<5 habitat trees	2
5-10 habitat trees	5
>10 habitat trees	7
> 10 native habitat trees	10

No. of understory and vegetation

<5% understory cover	0
5-15% understory cover	7
15-25% understory cover	13
>25% understory cover	19
>25% understory native cover	20

Environmental Weeds

No of action undertaken	
To identify or remove weeds	0
Weeds present, some action	
To manage weeds	8
No weeds present	15

Organic Litter ranking

<25% cover of organic litter	0
25-49% cover of organic litter	2
50-74% cover of organic litter	5
75-99% cover of organic litter	7
100% cover of organic litter	10

Logs and rocks

No logs and rocks	0
3-5 logs or rocks	3
Logs or rocks	5

Soil Management

No action plan for	
Soil managemen issues	0
Some attempt made to	
Manage soil management	
Issues	3
Soil manage effectively	5

Habitat Extras

No undelined	0
3 items underlined	3

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3-6 items underlined	10
7-10 items underlined	15
>10 items underlined	20

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Results and Discussions

Surfaces Within the University Ground

Table 1 presents the percentage cover of the surfaces within the campus. Soft surfaces (vegetation, and grassy play areas) covers much of the campus with 81.25 percentage cover. Buildings covers 47.92% followed by exotic trees with 37.58%, indigenous with 16.67%, exotic shrubs with 14.58%, hard impermeable surfaces (Basketball court, parking areas) with 12.5%, indigenous shrubs with 10.47%, native and indigenous grass cover with 6.25%, mulch cover with 6.25%, shrubs native with 0.41%, native exotic cover with 0.41% and orchards and gardens with also 0.41 percent cover.

Dyment, J. E (2008) states that an environmental factor of particular importance to children's physical activity levels appears to be the presence of parks and open space. Thus, in promoting children's health, school grounds merit consideration as a potential setting for intervention. This paper explores how 'green' the school grounds are, which contain a greater diversity of landscaping and design features, affect the quantity and quality of physical activity among elementary school children. Teachers, parents and administrators associated with 59 schools across Canada completed questionnaires (n = 105).

Analysis reveals that through greening school grounds, diversify the play repertoire, creates opportunities for students of all ages, interests and abilities to be more physically active. Seen in this light, green school grounds stand to be an important intervention to be included in school health promotion initiatives. Results of the study shows that the University grounds is mostly covered with soft surface. This means that there are rooms for biodeiversity improvement.

Table 1. Surfaces within the University ground

Surfaces	No. of boxes covered by surface/ha	Total number of boxes covering the school/ha	Percentage cover
Buildings	23	48	47.92
Hard Impermeable Surfaces	6	48	12.5
Soft Surfaces	39	48	81.25
Tress (Native)	3	48	4.16
Trees (Indigenous)	8	48	16.67
Trees (Exotic)	19	48	37.58
Shrubs (Native)	2	48	0.41

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Shrubs (Indigenous)	5	48	10.47
Shrubs (Exotic)	7	48	14.58
Native Grass Cover	3	48	6.25
Native Indigenous Cover	3	48	6.25
Native Exotic Cover	2	48	0.41
Orchards and food gardens	2	48	0.41
Mulch (no plant cover)	3	48	6.25

University School Ground Habitat Quality Assessment

Table 2 presented the assessment of the quality of habitat present in the school ground. Based on the result the overall score for the ground habitat quality is 67. For the number of trees the campus has more than trees per hectare with a habitat score of 10. For the number of habitat trees per hectare, the campus has more than 10 native habitat trees with a score of 10. For the understory and vegetation structure there were more than 25% understory ground cover with a habitat score of 19. For the environmental weeds there were weeds presents and some action were taken to manage weeds with a score of 8. For the organic litter, 25 – 49% cover of organic litter in gardens with a score of 2. For logs and rocks per hectare, >5 logs and rocks per hectare with a score of 5. Soil management issues present, some attempt made to manage these with a score of 3 and for the habitat extras more than 10 enhancements were underlined with a habitat score of 10.

According to Bail, S. M. (2017) due to invasive species and the assumption of an unhealthy ecosystem, Steidtmann Woods is an underutilized piece of land owned by Bowling Green State University. However, the property had never been analyzed. The proposed hypothesis was that Steidtmann was indeed in distress and its ecosystem could benefit from intervention—removal of invasive species and supplementation of natives. Through a rapid vegetative analysis, data was collected in several regions of the woods to identify what natives and non-natives were present as well as to identify their proportions. With that data, the species evenness, richness, and diversity was calculated. The data concluded that Steidtmann needs restoration and that the presence of invasive species reduces the properties overall health.

This study revealed that the Quality of habitat of Capsu Pilar Campus is very close the the aim score of 70 which is 67. Further, it is reflected that the area is habitable for both flora and fauna.

Tanle 2. Habitat Quality assessment of university school ground

Attribute	Description	Ranking	Score
Number of Trees/ha = 34.50 = <u>number of trees</u> number of hectares in the school	< 10 trees / ha		
	10 – 20 trees / ha		

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$= 122/3.536$ $=34.50$	> 20 trees / ha	X	10
Number of Habitat Trees/ha = 17.82 $= \frac{\text{number of habitat trees}}{\text{number of hectares in the school}}$ $=17.82/3.536$ $=17.82$	< 5 habitat trees/ha		
	5 – 10 habitat trees/ha		
	> 10 native habitat trees/ha	X	10
Understorey and Vegetation Structure <u>Underline when present:</u> Understorey: Shrub (1-5m), Small Shrub < 1m, Tall grass > 1m, Scrambler/climber Herb-layer: Fern, Moss/Lichen, Orchids, Native Percentage Cover of Understorey = Number of understorey boxes shaded in $\times 100$ $=13/48$ $=27.01$ * Number of boxes (in the grid) covering the school ground Grasses < 0.5m, Other _____	< 5% understorey cover in the school ground		
	5 –15% understorey cover in the school ground		
	15 – 25% understorey cover in the school ground		
	> 25% understorey cover in the school ground	X	19
	> 25% native understorey cover in the school ground		
Environmental Weeds	No action taken to identify or remove weeds		
	Weeds present - some action taken to manage weeds	X	8
	No weeds present		
Organic Litter = Organic litter includes leaves, twigs, small branches, tanbark and mulch < 30 cm circumference	< 25% cover of organic litter in gardens		
	25 – 49% cover of organic litter in gardens	x	2

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Percentage Cover of Organic Litter = $\frac{\text{Number of organic litter boxes shaded in}}{\text{* NSSEG}} \times 100$ = 18/39 = 0.729 x 100 = 72.90	50 – 74% cover of organic litter in gardens		
	75 – 99% cover of organic litter in gardens		
	100% cover of organic litter in gardens		
Logs and Rocks/ha = 339 = $\frac{\text{number of logs and rocks}}{\text{number of hectares in the school}}$ = 1200/3.536	No Logs or rocks		
	3 – 5 logs or rocks/ha		
	> 5 logs or rocks/ha	X	5
Soil Management	Soil management issues present, no action or plan to manage these made		
	Soil management issues present, some attempt made to manage these	X	3
	Soil managed effectively		
Habitat Extras <u>Underline when present:</u> <u>Composting area</u>, <u>Vegetable garden/ orchard</u>, <u>Lids on bins or bins located inside the buildings</u>, Indigenous plant propagation, <u>Plants in the classroom</u>, Plan to decrease rubbish in the school ground, Indigenous food garden, Plants (flora) in the school ground identified and labelled, Nest boxes, List of animals (fauna) in the school ground maintained, <u>Limited spraying of chemicals in the school ground</u>, <u>Fallen branches and leaf litter remaining in garden beds</u>, <u>Native plants naturally regenerating</u>, Other	0 enhancements underlined		
	< 3 enhancements underlined		
	3 – 6 enhancements underlined		
	7 – 10 enhancements underlined		
	> 10 enhancements underlined	X	10
TOTAL (out of 100)	Aim for score > 70		67

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Number of trees in the University Ground

Table 3 presents the number of trees found in the university ground. There were 122 trees recoded and identified. These were mahogany (56) considered as exotic and non habitat followed by Indian Mango (21) classified as Exotic and habitat tree, Rambutan (3) classified as exotic and a habitat tree, coconut tree (7) exotic and also classified as a habitat tree, Inyam (1) classified as native and habitable, Bitá (1) classified as native and habitable, Tipolo (1) classified as native and habitable, Gemilina (6) classified as Exotic and habitable, Talisay (8) classified as native and habitable, Santol (2) classified as native and habitable, Sampalok (1), classified as native and habitable, kamagong (6) classified as native and habitable, Narra tree (1) classified as native and habitable, Ipil-ipil (3) classified as native and habitable, Balete tree (3) classified as native and habitable, and guava tree (2), classified as native and habitable.

Dida, J. J. V., states that the University of the Philippines Los Baños (UPLB) campus is part of the Mount Makiling Forest Reserve (MMFR) buffer zone. It is home to diverse and unique species of plants and fauna. Given the important role of trees in sequestering atmospheric carbon, the estimation of aboveground biomass (AGB) will be useful in assessing green spaces and campus zoning in UPLB. Tree inventory activities were conducted in the campus to collect information such as geographic location and diameter at breast height of trees. Allometric equations developed for moist tropical zones were used to estimate the AGB of trees inside the campus. A total of 2,947 individual trees belonging to around 151 species were recorded, with *Swietenia macrophylla* as the most frequent species followed by *Pterocarpus indicus*. Tree species belonging to larger diameter class yielded higher aboveground biomass values compared to other diameter classes. The conduct of future inventory activities in the UPLB campus will aid in the monitoring of AGB yield and carbon stock through time

Data revealed that CapSU grounds grows diverse species of trees and they are present in a good number. This implies that the type of species is a potential habitat for local species.

Table 3. Number of trees in the university ground

Name of Trees	Number of trees	Native/Exotic/Indigenous	Habitat/Non habitat
Mahogany	56	Exotic	Non habitat
Indian Mango	21	Exotic	Habitat
Rambutan	3	Exotic	Habitat
Coconut tree	7	Exotic	Habitat
Inyam tree	1	Native	Habitat
Bitá	1	Native	Habitat
Tipolo	1	Native	Habitat
Gemilina	6	Exotic	Habitat
Talisay	8	Native	Habitat
Santol	2	Native	Habitat
Sampalok	1	Native	Habitat
Kamagong	6	Native	Habitat
Narra tree	1	Native	Habitat
Ipil-Ipil	3	Native	Habitat
Balete tree	3	Native	Habitat
Guava tree	2	Indigenous	Habitat
TOTAL	122		

Understory and vegetation within the campus

Table 4 presents the understory and vegetation type within the campus. Data showed that there were 10 understory and vegetation type within the campus. These were: Yellow Bell (25) classified as native; crotons (19) classified as exotic, bougainvillea (5) classified as native, pandakaki (30) classified as native, rose (2) classified as native, santan (6) classified as native, kugon (15) classified as indigenous, tanglad, (21) classified as indigenous, anahaw (5) classified as native and esopada-espada (37) classified as native.

Amoroso, V. B. (2021) states that the forest patches of Marilog District, Davao are the remnants of various anthropogenic activities including logging, conversion of land for agriculture, ecotourism and human settlements. Floristic study was carried out in 2018–2019 from 100 established plots measuring

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20 x 20 m, with repeated transect walks and opportunistic sampling along forest trails. One-hundred-and-four species of understory flowering plants were identified from 102 genera and 40 families. Species diversity mean values across study sites using Simpson's (D) and Shannon-Wiener index (H') were 0.97 and 3.9, respectively. Species diversity was highest in sites 2 and 4 (D = 0.98; H' = 4.0 each) and lowest in site 5 (D = 0.96; H' = 3.7). At family level, the most abundant taxa include Zingiberaceae (26 species) (15%), Orchidaceae (19 species) (11%), Gesneriaceae (14 species) (8%), and Rubiaceae and Arecaceae (13 species each) (7%). Conservation status assessment using International Union for Conservation of Nature (IUCN) revealed 10 threatened species, while listing from the Philippines' administrative order has categorized 13 threatened species. A total of 54 species (ca. 1.14% of the total Philippine endemic vascular flora) of understory flowering plants were Philippine endemics. Findings of this study were used as additional data for the proclamation of Mt. Malambo as Local Conservation Area, which was formalized through a barangay resolution.

Data collected from this study suggests diversified environment in the future. This includes insects such as butterflies, spiders and other related organism.

Table 4. Understory and vegetation type within the campus

Understory or Vegetation Type	Native/Exotic/Indigenous	No. found in school ground
1. Yellow Bell	Native	25
2. Crotons	Exotic	19
3. Bougainvillea	Native	5
4. Pandakaki	Native	30
5. Rose	Native	2
6. Santan	Native	6
7. Kugon	Indigenous	15
8. Tanglad	Indigenous	21
9. Anahaw	Native	5
10. Espada espada	Native	37
Total		137

Environmental Weeds within the Campus

Table 5 presents the environmental weeds within the campus. The data showed that there were 7 environmental weeds recorded and identified within the campus. These were carabao grass, amorsikos, mani-mani, makahiya, kataka-taka classified as indigenous cover and Bermuda grass classified as exotic cover. This implies that the place is habitable for flora and a potential place for laboratory activities of bio, agri and related studies.

Table 5. Environmental weeds within the campus

Name of Environmental Weeds	Native Cover/Exotic Cover/Indigenous Cover
1. Carabao Grass	Indigenous cover
2. Amorsikos	Indigenous cover
3. Mani-mani	Indigenous cover
4. Makahiya	Indigenous cover
5. Bermuda grass	Exotic cover
6. Tawa-tawa	Indigenous cover
7. Kataka-taka	Indigenous Cover

Fauna found in the campus

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Table 6 shows the fauna found within the campus. There were a total of 77 recorded and identified fauna belonging to 10 species. These were dragonfly (15), goat (3), cow (2), birds (5), spiders (3), ants (32), worms (6), frogs (2), dogs (3), and snails (6).

Pulhin, J. M. (2001) states that the present report is an initial attempt to look into the phenomenon of community participation in forest management and governance in the country, both within and outside the context of the government's CBFM program. It is the result of several months of literature review and limited site validation of documented and undocumented CBFM experiences. It seeks to characterize the various CBFM models that exist, identify their strengths and weaknesses, and articulate underlying issues that enhance or constrain community participation. In so doing, it hopes to serve as a building block for further research and, hopefully, future positive action that will secure the communities' access to their natural resource base, ensure the sustainability of their livelihood, and contribute to the increase in number of empowered individuals whose community organizations and institutions reinforce the sustainable use of their resources.

This implies that the place is habitable for fauna and a potential place for laboratory activities of bio, agri and related studies.

Name of Fauna	No. in the campus
1. Dragon fly	15
2. Goat	3
3. Cow	2
4. Birds	5
5. Spiders	3
6. Ants	32
7. Worms	6
8. Frogs	2
9. Dogs	3
10. Snails	6
Total	77

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusion

Much of the university ground were covered by soft surfaces (vegetation, play areas, park) and the quality of habitat assessment got a score of 67. There is a present of native trees recorded and most of the vegetation cover is native. Most of the environmental weeds were indigenous and some actions were taken to remove it. There is also 10 species of fauna identified and recorded in the campus. Therefore, the campus is a good site for tourism, conservation and instruction.

Recommendation

1. Plant more indigenous trees and indigenous plants around the campus for habitat of local flora and fauna.
2. Potential reference for policy on conservation of indigenous and native species is recommended to be crafted with in the campus as well as the municipality of Pilar for future tourism site.
3. This may also be a good start for promoting tourism and environmental conservations.

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(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.) Journal

Table Coffee Booklet or IEC materials

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

Dymont, J. E., & Bell, A. C. (2008). Grounds for movement: green school grounds as sites for promoting physical activity. *Health education research*, 23(6), 952-962.

Bail, S. M. (2017). Vegetative Rapid Assessment and Habitat Quality Analysis of Steidtmann Woods.

Dida, J. J. V., & Tiburan Jr, C. L. Aboveground biomass estimation of trees in the University of the Philippines Los Baños Campus, Philippines.

Acma, F. M., Mendez, N. P., Lagunday, N. E., & Amoroso, V. B. (2021). Diversity of understory flowering plants in the forest patches of Marilog District, Philippines. *Journal of Threatened Taxa*, 13(5), 18247-18256.

Guiang, E. S., Borlagdan, S. B., & Pulhin, J. M. (2001). Community-based forest management in the Philippines: a preliminary assessment. *Institute of Philippine Culture, Ateneo de Manila University*.